intro

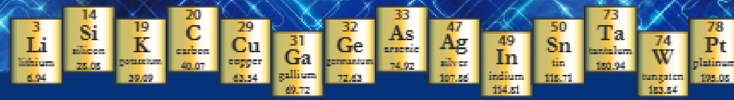
- china's role in critical mineral refining
    - The International Energy Agency reports that China refines about 90 percent of REEs and is the leading refiner for 19 of 20 key energy-related minerals, with an average global market share of around 70 percent.
    - What is striking is that for many high-demand minerals such as lithium, nickel, and cobalt, China’s share of raw extraction is only 10 to 30 percent, yet its share of refining and processing reaches 60 to 70 percent.
    - It also secured foreign mining assets in developing countries through the Belt and Road Initiative. Chinese-backed entities have invested nearly US$57 billion in mineral projects across 19 countries.23 This ensures raw material flows into China’s refining industry, reinforcing its role as the central hub of global mineral processing.

- where do minerals come from?
    - Australia is also a leading mining nation. In 2023, it supplied nearly half of global lithium and ranked among the top five producers of cobalt, manganese, and REEs.
    - Ukraine holds Europe’s largest reserves of lithium and titanium, as well as significant uranium, graphite, manganese, and REEs.36
    - South Africa is central to global supplies of manganese, chromite, vanadium, and platinum-group metals. It holds 80 percent of known manganese reserves and more than 70 percent of chromite. It is also a leading producer of vanadium and platinum.

mobile phone resources (2016)

- look:

USGS
science for a changing world



A World of Minerals in Your Mobile Device

Mobile phones and other high-technology communications devices could not exist without mineral commodities. More than one-half of all components in a mobile device—including its electronics, display, battery, speakers, and more—are made from mined and semi-processed materials (mineral commodities). Some mineral commodities can be recovered as byproducts during the production and processing of other commodities. As an example, bauxite is mined for its aluminum content, but gallium is recovered during the aluminum production process. The images below show the **ore minerals** (sources) of some mineral commodities that are used to make components of a mobile device. On the reverse side, the map and table depict the major source countries producing some of these mineral commodities along with how these commodities are used in mobile devices. For more information on minerals, visit http://minerals.usgs.gov.

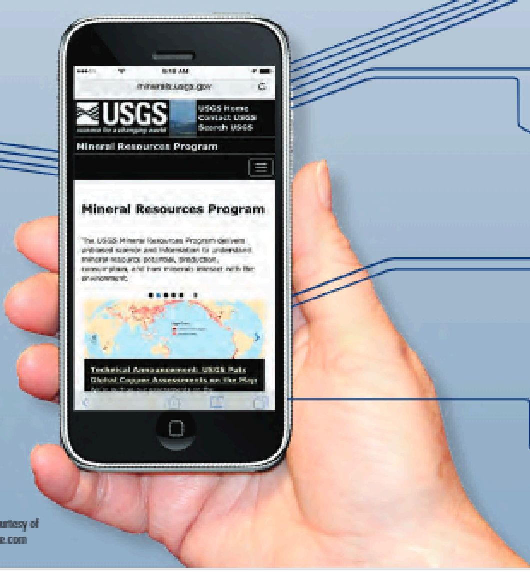
Display

A mobile device's glass screen is very durable because glassmakers combine its main ingredient, **silica** (silicon dioxide or quartz) **sand**, with ceramic materials and then add potassium.

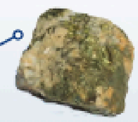
Layers of indium-tin-oxide are used to create transparent circuits in the display. Tin is also the ingredient in circuit board solder, and **cassiterite** is a primary source of tin.

Gallium provides light emitting diode (LED) backlighting. **Bauxite** is the primary source of this commodity.

Sphalerite is the source of indium (used in the screen's conductive coating) and germanium (used in displays and LEDs).



Electronics and Circuitry



The content of copper in a mobile device far exceeds the amount of any other metal. Copper conducts electricity and heat and comes from the source mineral **chalcopyrite**.

Tetrahedrite is a primary source of silver. Silver-based inks on composite boards create electrical pathways through a device.

Silicon, very abundant in the Earth's crust, is produced from the source mineral quartz and is the basis of integrated circuits.

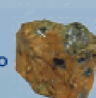
Arsenopyrite is a source of arsenic, which is used in radio frequency and power amplifiers.

Tantalum, from the source mineral **tantalite**, is added to capacitors to regulate voltage and improve the audio quality of a device.

Wolframite is a source of tungsten, which acts as a heat sink and provides the mass for mobile phone vibration.

Battery

Spodumene and subsurface brines are the sources of lithium used in cathodes of lithium-ion batteries.



Graphite is used for the anodes of lithium-ion batteries because of its electrical and thermal conductivity.

Speakers and Vibration

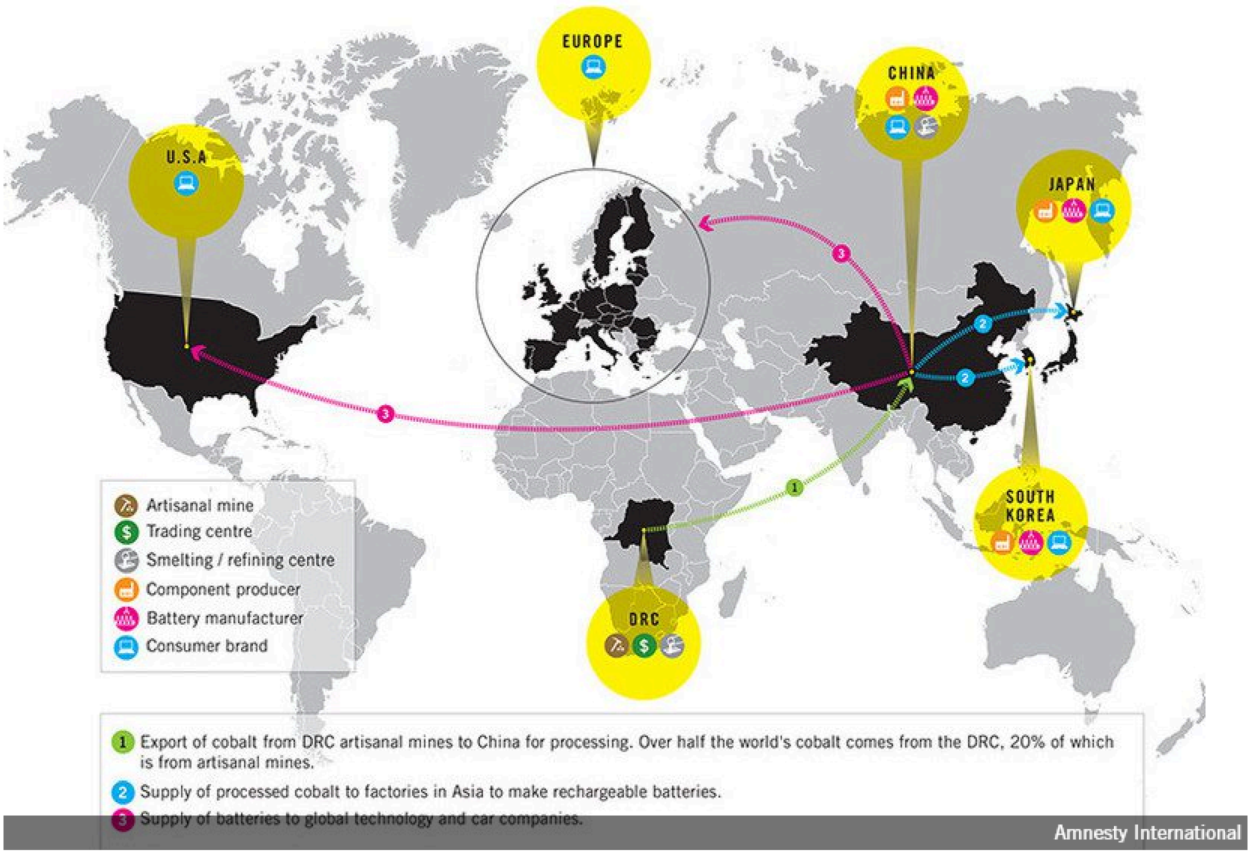
Bastnaesite is a source of rare-earth elements used to produce magnets in speakers, microphones, and vibration motors.

Banner image courtesy of freevector-archive.com

Element	Ores (minerals) occurring in the DRC	Selected uses of element
Copper (Cu)	A variety of Copper sulfide minerals	Electrical wire and cable; Electric motors; Integrated circuits
Cobalt (Co)	Heterogenite, a mixed oxide and Hydroxide of Cobalt ($CoOOH$)	Alloys for gas turbines and jet engines; Batteries in cell phones and Electric vehicles
Niobium (Nb)	Columbite (also called niobite), an Iron, Manganese, Niobium oxide mineral $[(Fe,Mn)Nb_2O_6]$	Alloys in jet and rocket engines; Superconducting magnets in magnetic resonance imaging equipment
Tantalum (Ta)	Tantalite, an Iron, Manganese, Tantalum oxide mineral $[(Fe,Mn)Ta_2O_6]$ Columbite-tantalite ores are typically referred to as coltan (see https://friendsofthecongo.org/coltan/)	Capacitors in cell phones and personal computers; Alloys in jet engines
Tin (Sn)	Cassiterite, a tin oxide mineral SnO_2	Solders for electronic circuits; Alloys with zirconium for cladding of nuclear fuel rods
Tungsten (W)	Wolframite an iron, manganese tungstate mineral $(Fe,Mn)WO_4$	Tungsten carbide for cutting tools; Alloys for use in armor- piercing ammunition
Gold (Au)	Native metal, Au	Monetary use; Jewelry; Electrical contacts in personal computers

cobalt

- byproduct of mining nickel and copper
- DRC - largest producer of cobalt in the world, 60% of world's supply
- 50% of the world's reserve of Cobalt
- 41% of cobalt demand driven by Batteries
- China - leading supplier of cobalt to the US but China gets its cobalt from Congo-Kinshasa

- blood batteries - [amnesty intl investigation coltan 2016]
- key mineral for green energy revolution - contributes to manufacture of more eco-friendly products
- 

How cobalt moves from mine to the global market

coltan

- coltan: columbite-tantalite, black-tar like mineral
- DRC - 64% world's reserves of coltan
- when refined, coltan becomes resistant powder that can hold a high electric charge

- vital element in creating devices that store energy / capacitors: mobile phones, computer chips, game consoles
- coltan mined by rebels within DRC and neighbors to DRC (Rwanda, Uganda, Burundi) sold to multi-national corporations, forming "coltan exploitation chain"

3TG: Tin, Tantalum, Tungsten, and Gold

- Tin (cassiterite), tantalum (coltan), tungsten (wolframite), and gold- commonly referred to as the “3TGs”- are the primary conflict minerals that drive continuous civil war in North and South Kivu.
- Together, these resources are necessary for many global industries and products, including smartphones and laptops, energy storage batteries, electric vehicles, artificial intelligence chips, and aerospace and military technologies.
- dozens of militia groups sought to control the lucrative 3TG (tin, tungsten, tantalum, gold) mineral territories in the eastern Congo. The

3TG minerals are vital components to consumer electronic devices and microprocessors.

DRC

- Despite these natural resources, the World Bank estimates that the DRC is among the five poorest nations in the world.3 In 2025, an estimated 74% of Congolese people lived on less than $2.15 per day.
- On 27 June 2025, the DRC and Rwanda signed a peace treaty, the Washington Accord, to end the conflict in the eastern DRC. The US is meant to provide security guarantees in the eastern Congo in exchange for access to vital mineral territories. Despite the signing, violence and insecurity in the area continues
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- MINING
    - Cobalt and copper, key components of green energy technology, are not always labeled as “conflict minerals,” but they play similar divisive roles in the DRC’s south-eastern “Copper Belt.”

- Fifteen of the 19 major copper-cobalt mining concessions in the DRC are majority-owned by Chinese mining companies